Empirically derived dietary patterns and risk of postmenopausal breast cancer in a large prospective cohort study.

BACKGROUND: Inconsistent associations have been reported between diet and breast cancer. OBJECTIVE: We prospectively examined the association between dietary patterns and postmenopausal breast cancer risk in a US-wide cohort study. DESIGN: Data were analyzed from 40 559 women who completed a self-administered 61-item Block food-frequency questionnaire in the Breast Cancer Detection Demonstration Project, 1987-1998; 1868 of those women developed breast cancer. Dietary patterns were defined by using principal components factor analysis. Cox proportional hazard regression was used to assess breast cancer risk. RESULTS: Three major dietary patterns emerged: vegetable-fish/poultry-fruit, beef/pork-starch, and traditional southern. The vegetable-fish/poultry-fruit pattern was associated with higher education than were the other patterns, but was similar in nutrient intake to the traditional southern pattern. After adjustment for confounders, there was no significant association between the vegetable-fish/poultry-fruit and beef/pork-starch patterns and breast cancer. The traditional southern pattern, however, was associated with a nonsignificantly reduced breast cancer risk among all cases (in situ and invasive) that was significant for invasive breast cancer (relative hazard = 0.78; 95% CI = 0.65, 0.95; P for trend = 0.003). This diet was also associated with a reduced risk in women without a family history of breast cancer (P = 0.05), who were underweight or normal weight [body mass index (in kg/m(2)) < 25; P = 0.02], or who had tumors positive for estrogen receptor (P = 0.01) or progesterone receptor (P = 0.003). Foods in the traditional southern pattern associated with reduced breast cancer risk were legumes, low mayonnaise-salad dressing intake, and possibly cabbage. CONCLUSIONS: The traditional southern diet or its components are associated with a reduced risk of invasive breast cancer in postmenopausal women.